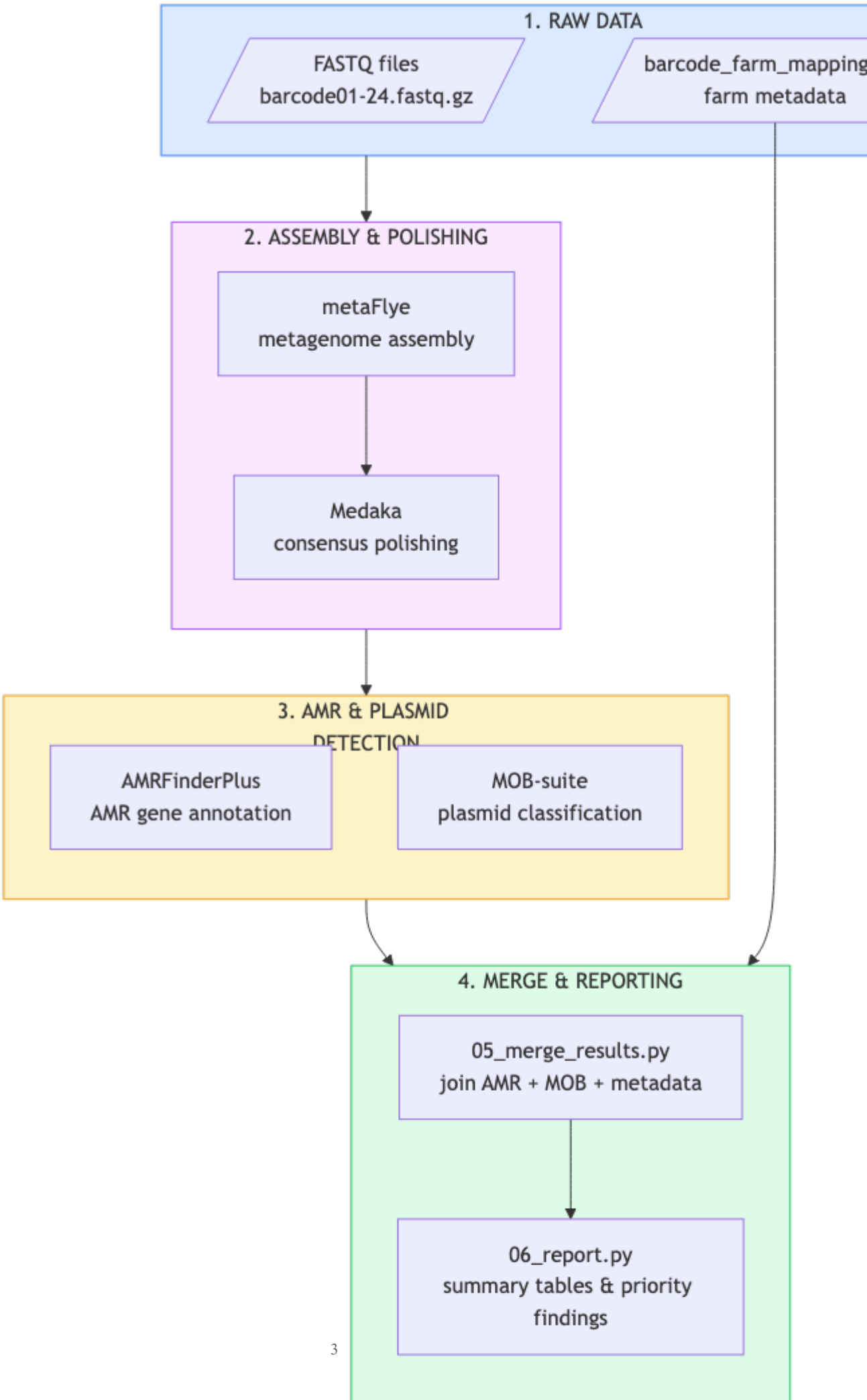
PHAGENT_AMR Workflow Documentation

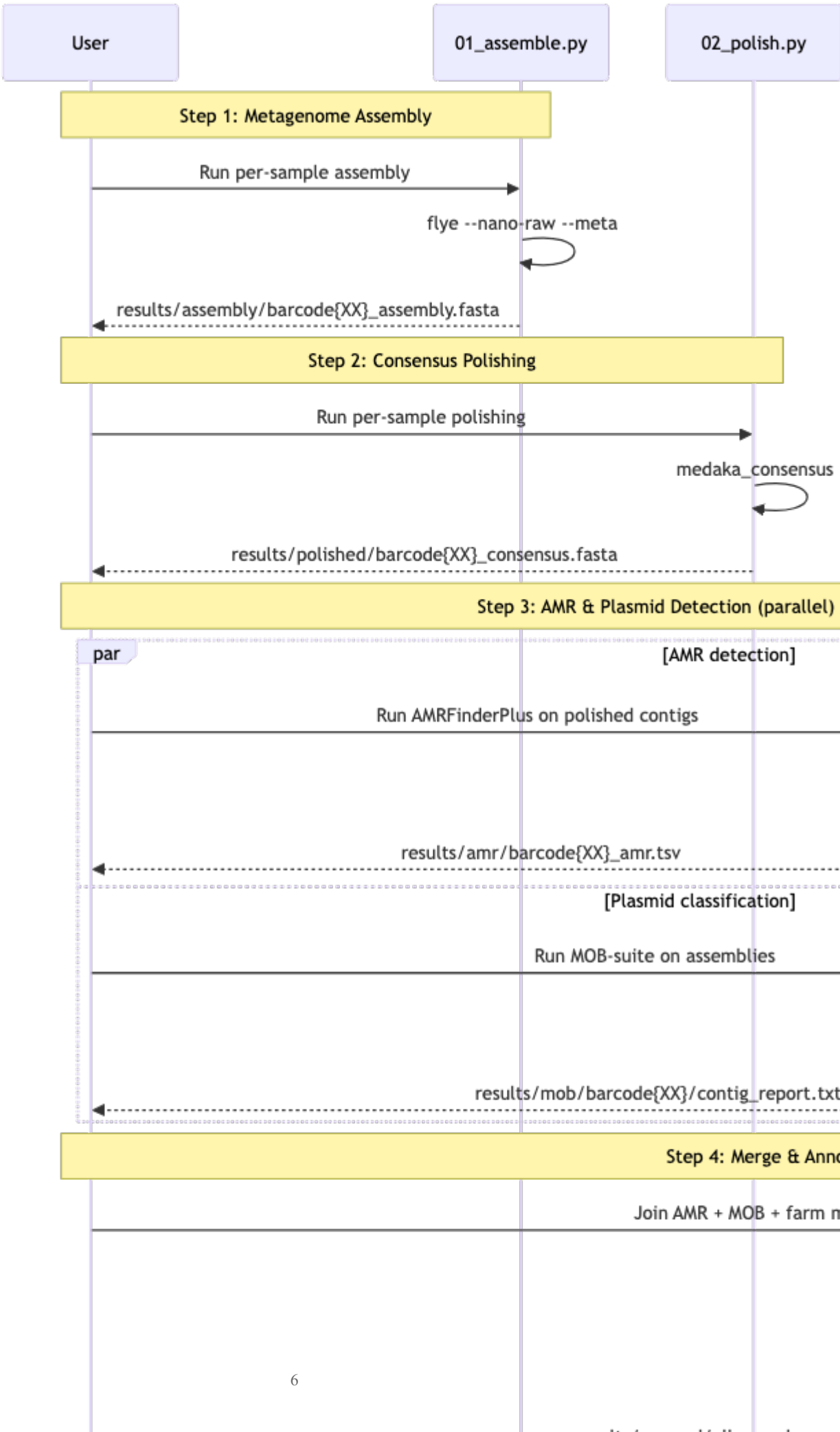
This document provides visual documentation of the assembly-based AMR profiling pipeline.

Pipeline Overview


1. RAW DATA
FASTQ files
barcode01-24.fastq.gz
barcode_farm_mapping
farm metadata
2. ASSEMBLY & POLISHING
metaFlye
metagenome assembly
Medaka
consensus polishing
3. AMR & PLASMID
DETECTION
AMRFinderPlus
AMR gene annotation
MOB-suite
plasmid classification
4. MERGE & REPORTING
05_merge_results.py
join AMR + MOB + metadata
06_report.py
summary tables & priority
findings

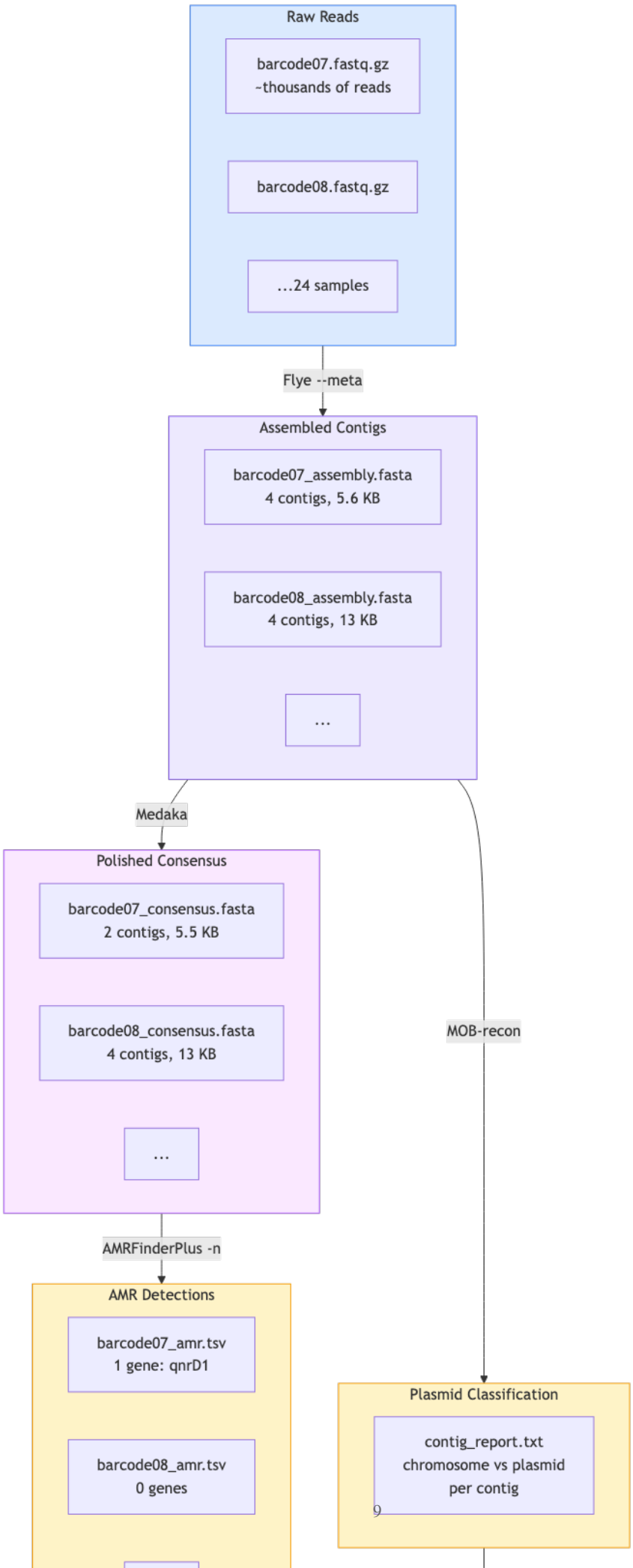
Pipeline Overview

Data Flow Sequence


User
01_assemble.py
02_polish.py
Step 1: Metagenome Assembly
Run per-sample assembly
flye --nano-raw --meta
results/assembly/barcode{XX}_assembly.fasta
Step 2: Consensus Polishing
Run per-sample polishing
medaka_consensus
results/polished/barcode{XX}_consensus.fasta
Step 3: AMR & Plasmid Detection (parallel)
par
[AMR detection]
Run AMRFinderPlus on polished contigs
results/amr/barcode{XX}_amr.tsv
[Plasmid classification]
Run MOB-suite on assemblies
results/mob/barcode{XX}/contig_report.txt
Step 4: Merge & Anno
Join AMR + MOB + farm m

Data Flow Sequence

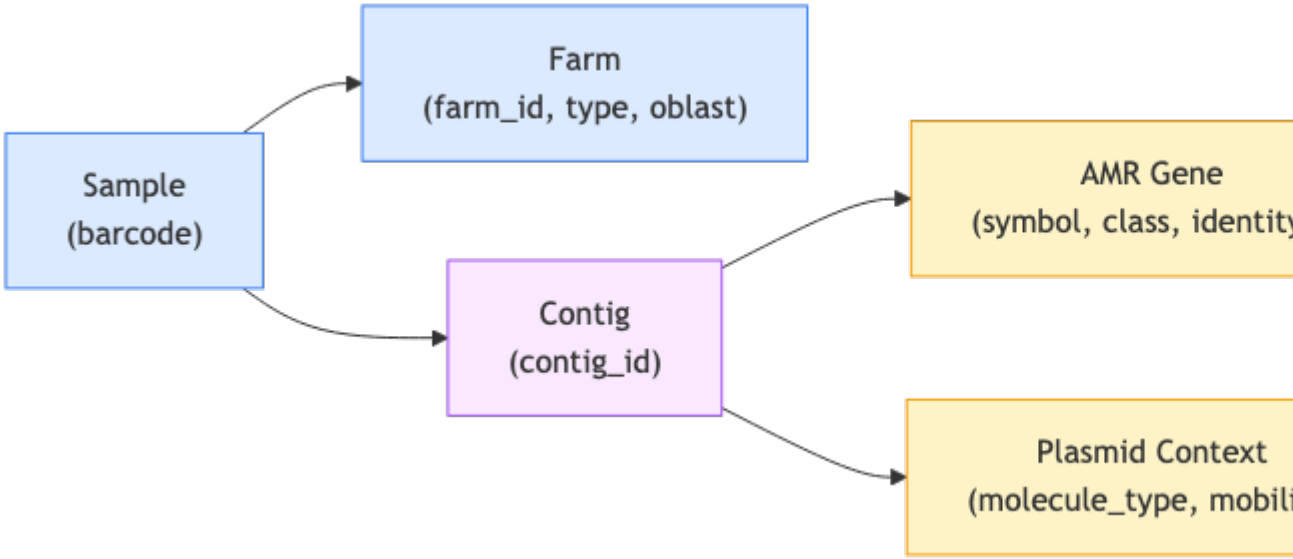
Data Transformation Diagram


Raw Reads
barcode07.fastq.gz
~thousands of reads
barcode08.fastq.gz
...24 samples
Flye --meta
Assembled Contigs
barcode07_assembly.fasta
4 contigs, 5.6 KB
barcode08_assembly.fasta
4 contigs, 13 KB
...
Medaka
MOB-recon
Polished Consensus
barcode07_consensus.fasta
2 contigs, 5.5 KB
barcode08_consensus.fasta
4 contigs, 13 KB
...
AMRFinderPlus -n
AMR Detections
barcode07_amr.tsv
1 gene: qnrD1
barcode08_amr.tsv
0 genes
Plasmid Classification
contig_report.txt
chromosome vs plasmid
per contig

Data Transformation

Evidence Model

Each AMR finding is traceable through the full chain:



Evidence Model

Tool Chain Summary

Step	Tool	Input	Output	Key Parameters
Assembly	metaFlye	Raw FASTQ	Contigs FASTA	`--nano-raw --meta`
Polishing	Medaka	FASTQ + Assembly	Consensus FASTA	`-m r1041_e82_400bps_sup_v5.2.0`
AMR Detection	AMRFinderPlus	Polished contigs	AMR gene TSV	`-n` (nucleotide mode)
Plasmid Typing	MOB-suite	Assembly contigs	Contig report	`mob_recon`
Merge	Python/pandas	AMR + MOB + metadata	Annotated table	Left joins on barcode + contig

Step	Tool	Input	Output	Key Parameters
Report	Python/pandas	Annotated table	Summary CSVs	Priority filtering, aggregation

---

Comparison with Main Pipeline

Aspect	Main Pipeline	PHAGENT_AMR
AMR source	ResFinder (read-level)	AMRFinderPlus (contig-level)
Evidence type	Individual reads	Assembled contigs
Plasmid context	None	MOB-suite classification
Confidence	Coverage % + Identity %	Assembly + gene coordinates
Scope	All 24 barcodes	Currently 4 (scalable to 24)
Best for	Broad screening	High-confidence, evidence-backed findings

---